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Electrostatics

NEET Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	74 questions
Answer key	at the end

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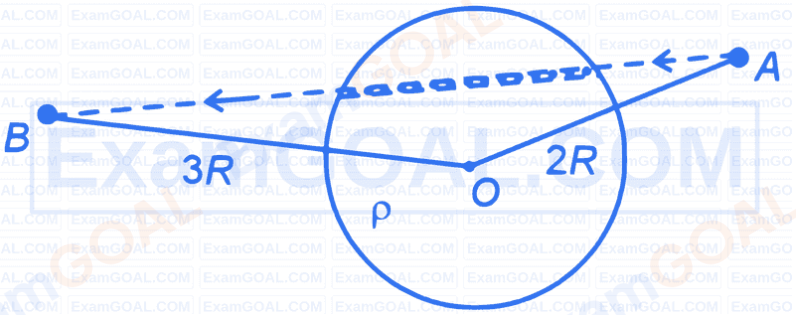
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MCQ (Single Correct Answer) 74 questions

Q1. RE-NEET 2026

Open solution

A unit positive point charge is taken slowly through an infinitesimally thin tube that is inside a charged dielectric sphere of radius R, having uniform positive charge density ρ, as shown in the figure. The initial and final positions of the charge are marked by A and B at distance $2R$ and $3R$ respectively, from the centre of the sphere. In this process, the magnitude of the total work done on the point charge is $\frac{\rho R^2}{n\varepsilon_0}$. The value of n is : (ε_0 is the permittivity of vacuum)



(A) 18 (B) 2 (C) 6 (D) 9

Q2. RE-NEET 2026

Open solution

Consider a fixed uniformly charged insulating sphere with radius R and total charge $+Q$. A point charge $-q$ ($q \ll Q$) with mass m is released from rest at a distance of $3R$ from the centre of the charged sphere. When the point charge reaches the surface of the sphere, its speed is:

(ε_0 is the permittivity of vacuum, neglect gravitational forces).

(A) $\sqrt{\frac{Qq}{4\pi\varepsilon_0 mR}}$ (B) $\sqrt{\frac{3Qq}{4\pi\varepsilon_0 mR}}$

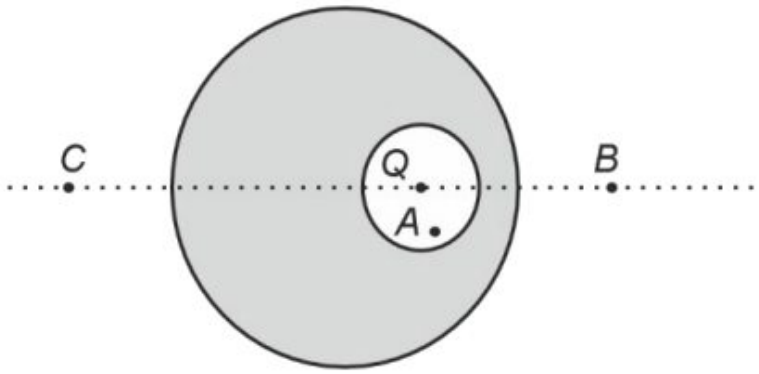
(C) $\sqrt{\frac{2Qq}{3\pi\varepsilon_0 mR}}$ (D) $\sqrt{\frac{Qq}{3\pi\varepsilon_0 mR}}$

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Q3. RE-NEET 2026

Open solution

A point charge Q is placed inside a cavity within a solid isolated conducting sphere. Consider points A, B and C as shown in the figure, where the magnitudes of the electric fields are E_A, E_B, E_C, respectively. The points B and C are at the same distance from the center of the solid sphere. The correct option is :



(A) $E_A \neq 0, E_B < E_C$ (B) $E_A = 0, E_B = E_C$ (C) $E_A \neq 0, E_B = E_C$ (D) $E_A = 0, E_B > E_C$

Q4. NEET 2026

Open solution

Which of the following statements are correct?

A. Inside a conductor, the electrostatic field is zero.

B. Electric field at the surface of a charged conductor does not depend on its surface charge density.

C. The interior of a charged conductor can have no excess charge in the static situation.

D. At the surface of a charged conductor, the electrostatic field must be normal to the surface at every point.

E. The electrostatic potential is zero everywhere inside a charged conductor.

Choose the correct answer from the options given below:

(A) A, B and D only (B) A, C and E only (C) A, C and D only (D) C, D and E only

Q5. NEET 2025

Open solution

An electric dipole with dipole moment $5 \times 10^{-6}\text{Cm}$ is aligned with the direction of a uniform electric field of magnitude $4 \times 10^5\ \text{N/C}$. The dipole is then rotated through an angle of 60° with respect to the electric field. The change in the potential energy of the dipole is:

(A) 1.2 J (B) 1.5 J (C) 0.8 J (D) 1.0 J

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Q6. NEET 2025

Two identical charged conducting spheres A and B have their centres separated by a certain distance. Charge on each sphere is q and the force of repulsion between them is F. A third identical uncharged conducting sphere is brought in contact with sphere A first and then with B and finally removed from both. New force of repulsion between spheres A and B (Radii of A and B are negligible compared to the distance of separation so that for calculating force between them they can be considered as point charges) is best given as:

Open solution

(A) $\frac{F}{2}$ (B) $\frac{3F}{8}$ (C) $\frac{3F}{5}$ (D) $\frac{2F}{3}$

Q7. NEET 2024 (Re-Examination)

A metal cube of side $5\ \text{cm}$ is charged with $6\mu\text{C}$. The surface charge density on the cube is

Open solution

(A) $0.125 \times 10^{-3}\text{Cm}^{-2}$ (B) $0.25 \times 10^{-3}\text{Cm}^{-2}$

(C) $4 \times 10^{-3}\text{Cm}^{-2}$ (D) $0.4 \times 10^{-3}\text{Cm}^{-2}$

Q8. NEET 2024 (Re-Examination)

The value of electric potential at a distance of $9\ \text{cm}$ from the point charge $4 \times 10^{-7}\text{C}$ is [Given $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ \text{N m}^2\text{C}^{-2}$] :

Open solution

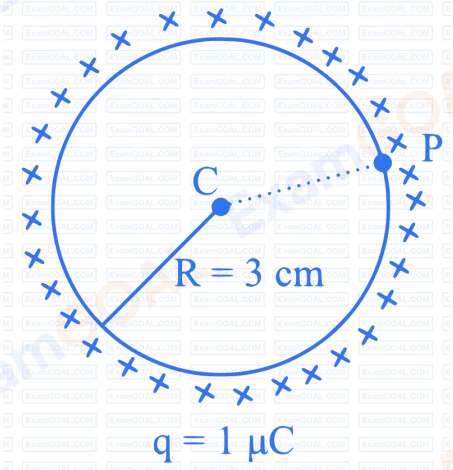
(A) $4 \times 10^2\ \text{V}$ (B) $44.4\ \text{V}$ (C) $4.4 \times 10^5\ \text{V}$ (D) $4 \times 10^4\ \text{V}$

Q9. NEET 2024

A thin spherical shell is charged by some source. The potential difference between the two points C and P (in V) shown in the figure is:

Open solution

(Take $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ SI units)



(A) 3×10^5 (B) 1×10^5 (C) 0.5×10^5 (D) Zero

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Q10. NEET 2024

Open solution

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: The potential (V) at any axial point, at $2\,\mathrm{m}$ distance (r) from the centre of the dipole of dipole moment vector $\vec{P}$ of magnitude, $4 \times 10^{-6}\mathrm{Cm}$, is $\pm 9 \times 10^{3}\ \mathrm{V}$.

(Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^{9}\mathrm{SI}$ units)

Reason R: $V = \pm\frac{2P}{4\pi\epsilon_0 r^2}$, where r is the distance of any axial point, situated at $2\,\mathrm{m}$ from the centre of the dipole.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true and R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Q11. NEET 2023 Manipur

Open solution

According to Gauss law of electrostatics, electric flux through a closed surface depends on :

(A) the area of the surface

(B) the quantity of charges enclosed by the surface

(C) the shape of the surface

(D) the volume enclosed by the surface

Q12. NEET 2023 Manipur

Open solution

A charge $\mathrm{Q}\ \mu\mathrm{C}$ is placed at the centre of a cube. The flux coming out from any one of its faces will be (in SI unit) :

(A) $\frac{Q}{\epsilon_0} \times 10^{-6}$

(B) $\frac{2\mathrm{Q}}{3\epsilon_0} \times 10^{-3}$

(C) $\frac{\mathrm{Q}}{6\epsilon_0} \times 10^{-3}$

(D) $\frac{\mathrm{Q}}{6\epsilon_0} \times 10^{-6}$

Q13. NEET 2023 Manipur

Open solution

If a conducting sphere of radius R is charged. Then the electric field at a distance $\mathrm{r}(\mathrm{r} > \mathrm{R})$ from the centre of the sphere would be, $(\mathrm{V} =$ potential on the surface of the sphere)

(A) $\frac{rV}{R^2}$

(B) $\frac{R^2V}{r^3}$

(C) $\frac{RV}{r^2}$

(D) $\frac{\mathrm{V}}{\mathrm{r}}$

Q14. NEET 2023

Open solution

An electric dipole is placed at an angle of 30° with an electric field of intensity $2 \times 10^{5}\mathrm{NC}^{-1}$. It experiences a torque equal to $4\ \mathrm{N\,m}$. Calculate the magnitude of charge on the dipole, if the dipole length is $2\ \mathrm{cm}$.

(A) 6 mC

(B) 4 mC

(C) 2 mC

(D) 8 mC

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Q15. NEET 2023

Open solution

If $\oint_s \vec{E} \cdot \overrightarrow{dS} = 0$ over a surface, then:

(A) the magnitude of electric field on the surface is constant.

(B) all the charges must necessarily be inside the surface.

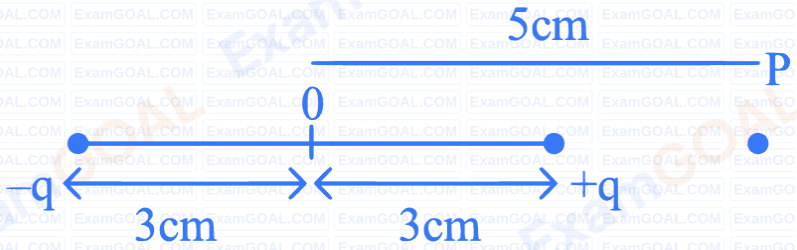
(C) the electric field inside the surface is necessarily uniform.

(D) the number of flux lines entering the surface must be equal to the number of flux lines leaving it.

Q16. NEET 2023

Open solution

An electric dipole is placed as shown in the figure.



The electric potential (in 10^2 V) at point P due to the dipole is ($\in_0$ = permittivity of free space and $\frac{1}{4\pi\epsilon_0}$ = K) :

(A) $\left(\frac{5}{8}\right)$qK

(B) $\left(\frac{8}{5}\right)$qK

(C) $\left(\frac{8}{3}\right)$qK

(D) $\left(\frac{3}{8}\right)$qK

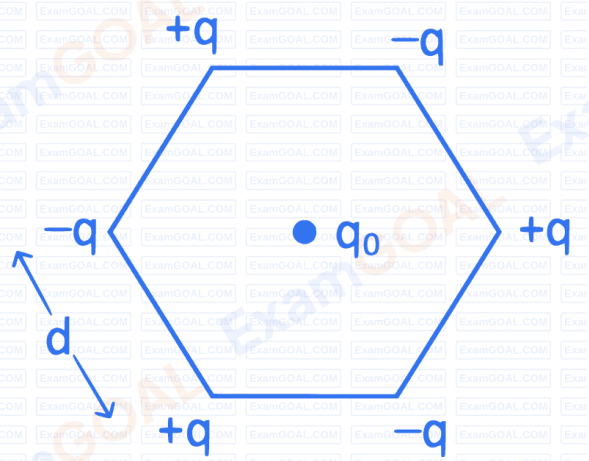
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Q17. NEET 2022 Phase 2

Six charges +q, −q, +q, −q, +q, and −q are fixed at the corners of a hexagon of side d as shown in the figure. The work done in bringing a charge q_0 to the centre of the hexagon from infinity is

(ε_0 - permittivity of free space)

Open solution



(A) $\frac{-q^2}{4\pi\varepsilon_0 d}\left(6-\frac{1}{\sqrt{2}}\right)$

(B) Zero

(C) $\frac{-q^2}{4\pi\varepsilon_0 d}$

(D) $\frac{-q^2}{4\pi\varepsilon_0 d}\left(3-\frac{1}{\sqrt{2}}\right)$

Q18. NEET 2022 Phase 1

The angle between the electric lines of force and the equipotential surface is

(A) 0° (B) 45° (C) 90° (D) 180°

Open solution

Q19. NEET 2022 Phase 1

Two hollow conducting spheres of radii R_1 and R_2 ($R_1 >> R_2$) have equal charges. The potential would be

Open solution

(A) More on bigger sphere

(B) More on smaller sphere

(C) Equal on both the spheres

(D) Dependent on the material property of the sphere

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Q20. NEET 2022 Phase 1

Two point charges −q and +q are placed at a distance of L, as shown in the figure.



Open solution

The magnitude of electric field intensity at a distance R(R >> L) varies as:

(A) $\frac{1}{R^2}$ (B) $\frac{1}{R^3}$ (C) $\frac{1}{R^4}$ (D) $\frac{1}{R^6}$

Q21. NEET 2021

Polar molecules are the molecules :

(A) having a permanent electric dipole moment

(B) having zero dipole moment

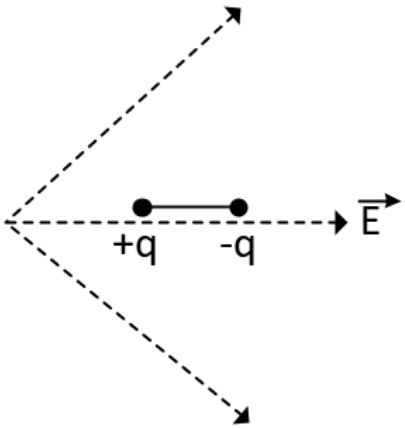
(C) acquire a dipole moment only in the presence of electric field due to displacement of charges

(D) acquire a dipole moment only when magnetic field is absent

Open solution

Q22. NEET 2021

A dipole is placed in an electric field as shown. In which direction will it move?



Open solution

(A) towards the right as its potential energy will increase.

(B) towards the left as its potential energy will increase.

(C) towards the right as its potential energy will decrease.

(D) towards the left as its potential energy will decrease.

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Q23. NEET 2021

Two charged spherical conductors of radius R_1 and R_2 are connected by a wire. Then the ratio of surface charge densities of the spheres (σ_1 / σ_2) is :

(A) $\frac{R_1^2}{R_2^2}$
(B) $\frac{R_1}{R_2}$
(C) $\frac{R_2}{R_1}$
(D) $\sqrt{\left(\frac{R_1}{R_2}\right)}$

Open solution

Q24. NEET 2021

Twenty seven drops of same size are charged at 220V each. They combine to form a bigger drop. Calculate the potential of the bigger drop.

(A) 1980 V (B) 660 V (C) 1320 V (D) 1520 V

Open solution

Q25. NEET 2020 Phase 1

A short electric dipole has a dipole moment of 16×10^{-9} Cm. The electric potential due to the dipole at a point at a distance of 0.6 m from the centre of the dipole, situated on a line making an angle of 60° with the dipole axis is :

$\left(\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 Nm^2/C^2\right)$

(A) 200 V (B) 400 V (C) zero (D) 50 V

Open solution

Q26. NEET 2020 Phase 1

In a certain region of space with volume 0.2 m^3, the electric potential is found to be 5 V throughout. The magnitude of electric field in this region is :

(A) 0.5 N/C (B) 1 N/C (C) 5 N/C (D) zero

Open solution

Q27. NEET 2020 Phase 1

A spherical conductor of radius 10 cm has a charge of 3.2×10^{-7} C distributed uniformly. What is the magnitude of electric field at a point 15 cm from the centre of the sphere?

$\left(\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 Nm^2/c^2\right)$

(A) 1.28×10^5 N/C
(B) 1.28×10^6 N/C
(C) 1.28×10^7 N/C
(D) 1.28×10^4 N/C

Open solution

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Q28. NEET 2019

A hollow metal sphere of radius R is uniformly charged. The electric field due to the sphere at a distance r from the centre :

(A) zero as r increases for r < R, decreases as r increases for r > R

(B) zero as r increases for r < R, increases as r increases for r > R.

(C) decreases as r increases for r < R and for r > R.

(D) increases as r increases for r < R and for r > R.

Open solution

Q29. NEET 2019

Two point charges A and B, having charges +Q and – Q respectively, are placed at certain distance apart and force acting between them is F. If 25% charge of A is transferred to B, then force between the charges becomes :

(A) F (B) $\frac{16F}{9}$ (C) $\frac{9F}{16}$ (D) $\frac{4F}{3}$

Open solution

Q30. NEET 2019

Two parallel infinite line charges with linear charge densities +λ C/m and -λ C/m are placed at a distance of 2R in free space. What is the electric field mid-way between the two line charges?

(A) $\frac{2\lambda}{\pi\varepsilon_0 R}$ N/C (B) zero

(C) $\frac{\lambda}{\pi\varepsilon_0 R}$ N/C (D) $\frac{\lambda}{2\pi\varepsilon_0 R}$ N/C

Open solution

Q31. NEET 2018

An electron falls from rest through a vertical distance h in a uniform and vertically upward directed electric field E. The direction of electric field is now reversed, keeping its magnitude the same. A proton is allowed to fall from rest in it through the same vertical distance h. The time of fall of the electron, in comparison to the time of fall of the proton is

(A) smaller (B) 5 times greater (C) 10 times greater (D) equal

Open solution

Q32. NEET 2017

Suppose the charge of a proton and an electron differ slightly. One of them is $-e$, the other is (e + Δe). If the net of electrostatic force and gravitational force between two hydrogen atoms placed at a distance d (musch greater than atomic size) apart is zero, then Δe is of the order of

[Given : mass of hydrogen $m_h = 1.67 \times 10^{-27}$ kg]

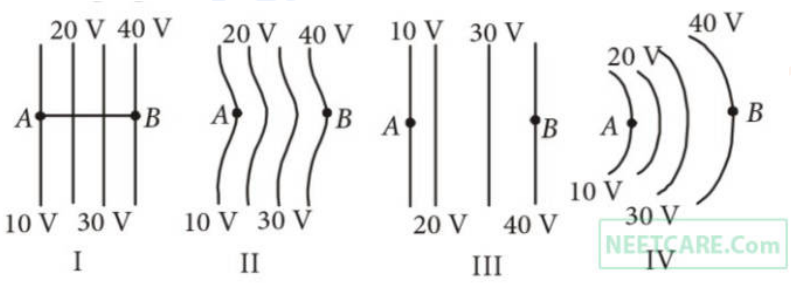
(A) 10^{-23} C (B) 10^{-37} C (C) 10^{-47} C (D) 10^{-20} C

Open solution

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Q33. NEET 2017

The diagrams below show regions of equipotentials.



A positive charge is moved from A to B in each diagram.

(A) In all the four cases the work done is the same.

(B) Minimum work is required to move q in figure(I).

(C) Maximum work is required to move q in figure (II).

(D) Maximum work is required to move q in figure (III)

Open solution

Q34. NEET 2016 Phase 2

An electric dipole is placed at an angle of 30° with an electric field intensity 2×10^{5} N C^{-1}. It experiences a torque equal to 4 N m. The charge on the dipole, if the dipole length is 2 cm, is

(A) 8 mC (B) 2 mC (C) 5 mC (D) 7 μC

Open solution

Q35. NEET 2016 Phase 1

Two identical charged spheres suspended from a common point by two massless strings of lengths l, are initially at a distance d(d < < l) apart because of their mutual repulsion. The charges begin to leak from both the spheres at a constant rate. As a result, the spheres approach each other with a velocity v. Then v varies as a function of the distance x between the spheres, as

(A) $v \propto x^{-1/2}$ (B) $v \propto x^{-1}$ (C) $v \propto x^{1/2}$ (D) $v \propto x$

Open solution

Q36. AIPMT 2015

If potential (in volts) in a region is expressed as V(x, y, z) = 6xy $-$ y + 2yz, the electric field (in N/C) at point (1, 1, 0) is

(A) $-\left(2\hat{i}+3\hat{j}+\hat{k}\right)$ (B) $-\left(6\hat{i}+9\hat{j}+\hat{k}\right)$

(C) $-\left(3\hat{i}+5\hat{j}+3\hat{k}\right)$ (D) $-\left(6\hat{i}+5\hat{j}+2\hat{k}\right)$

Open solution

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Q37. AIPMT 2015 Cancelled Paper

The electric field in a certain region is acting radially outward and is given by E = Ar. A charge contained in a sphere of radius 'a' centred at the origin of the field, will be given by

(A) $4\pi\varepsilon_0 Aa^3$ (B) $\varepsilon_0 Aa^3$ (C) $4\pi\varepsilon_0 Aa^2$ (D) $A\varepsilon_0 a^2$

Open solution

Q38. AIPMT 2014

A conducting sphere of radius R is given a charge Q. The electric potential and the electric field at the centre of the sphere rrespectively are

(A) zero and $\frac{Q}{4\pi\varepsilon_0 R^2}$

(B) $\frac{Q}{4\pi\varepsilon_0 R}$ and zero

(C) $\frac{Q}{4\pi\varepsilon_0 R}$ and $\frac{Q}{4\pi\varepsilon_0 R^2}$

(D) both are zero

Open solution

Q39. AIPMT 2014

In a region, the potential is represented by V(x, y, z) = 6x − 8xy − 8y + 6yz, where V is in volts and x, y, z are in metres. The electric force experienced by a charge of 2 coulomb situated at point (1, 1, 1) is

(A) $6\sqrt{5}$ N (B) 30 N (C) 24 N (D) $4\sqrt{35}$ N

Open solution

Q40. NEET 2013 (Karnataka)

An electric dipole of dipole moment p is aligned parallel to a uniform electric field E. The energy required to rotate the dipole by 90° is

(A) p^2 E (B) pE (C) infinity (D) pE^2

Open solution

Q41. NEET 2013 (Karnataka)

A charge q is placed at the centre of the line joining two equal charges Q. The system of the three charges will be in equilibrium if q is equal to

(A) −Q/4 (B) Q/4 (C) −Q/2 (D) Q/2

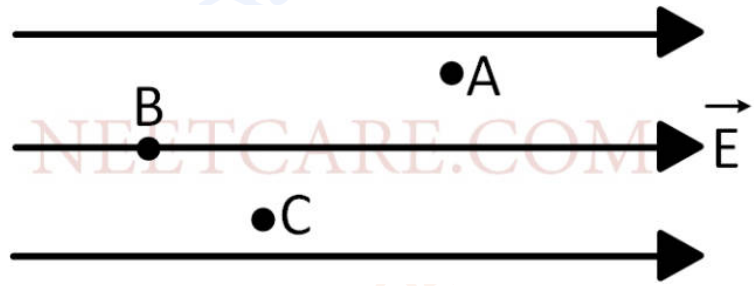
Open solution

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Q42. NEET 2013

A, B and C are three points in a uniform electric field. The electric potential is

Open solution

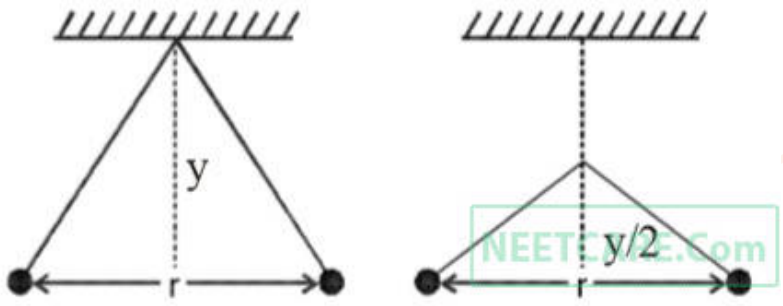


(A) maximum at C
(B) same at all the three points A, B and C
(C) maximum at A
(D) maximum at B

Q43. NEET 2013

Two pith balls carrying equal charges are suspended from a common point by strings of equal length, the equilibrium separation between them is r. Now the strings are rigidly clamped at half the height. The equilibrium separation between the balls now becomes

Open solution



(A) $\left(\frac{2r}{\sqrt{3}}\right)$
(B) $\left(\frac{2r}{3}\right)$
(C) $\left(\frac{1}{\sqrt{2}}\right)^2$
(D) $\left(\frac{r}{\sqrt[3]{2}}\right)$

Q44. AIPMT 2012 Mains

Two metallic spheres of radii 1 cm and 3 cm are given charges of -1×10^{-2} C and 5×10^{-2} C, respectively. If these are connected by a conducting wire, the final charge on the bigger sphere is

Open solution

(A) 2×10^{-2} C
(B) 3×10^{-2} C
(C) 4×10^{-2} C
(D) 1×10^{-2} C

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Q45. AIPMT 2012 Prelims

An electric dipole of moment p is placed in an electric field of intensity E. The dipole acquires a position such that the axis of the dipole makes an angle θ with the direction of the field. Assuming that the potential energy of the dipole to be zero when $\theta = 90^{o}$, the torque and the potential energy of the dipole will respectively be

Open solution

(A) pEsinθ, $-$pEcosθ

(B) pEsinθ, -2pEcosθ

(C) pEsinθ, 2pEcosθ

(D) pEcosθ, $-$pEsinθ

Q46. AIPMT 2012 Prelims

Four point charges $-$Q, $-$q, 2q and 2Q are placed, one at each corner of the square. The relation between Q and q for which the potential at the centre of the square is zero is

Open solution

(A) Q = $-$q

(B) Q = $-\frac{1}{q}$

(C) Q = q

(D) Q = $\frac{1}{q}$

Q47. AIPMT 2012 Prelims

What is the flux through a cube of side a if a point charge of q is at one of its corners?

Open solution

(A) $\frac{2q}{\varepsilon_0}$

(B) $\frac{q}{8\varepsilon_0}$

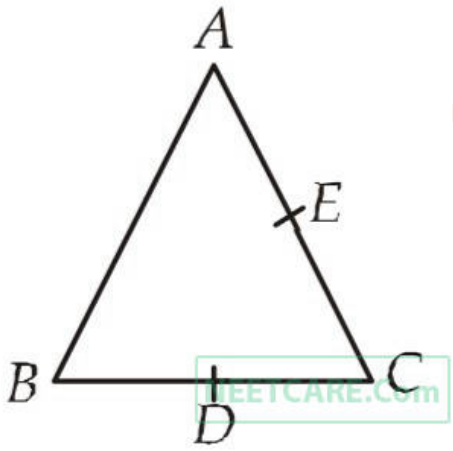
(C) $\frac{q}{\varepsilon_0}$

(D) $\frac{q}{2\varepsilon_0}6a^2$

Q48. AIPMT 2011 Mains

Three charges, each +q, are placed at the corners of an isosceles triangle ABC of sides BC and AC, $2a$. D and E are the mid points of BC and CA. The work done in taking a charge Q from D to E is

Open solution



(A) $\frac{3qQ}{4\pi\varepsilon_0 a}$

(B) $\frac{3qQ}{8\pi\varepsilon_0 a}$

(C) $\frac{qQ}{4\pi\varepsilon_0 a}$

(D) zero

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Q49. AIPMT 2011 Mains

The electric potential V at any point (x, y, z), all in metres in space is given by $V = 4x^2$ volt. The electric field at the point (1, 0, 2) in volt/meter, is

(A) 8 along negative X-axis
(B) 8 along positive X-axis
(C) 16 along negative X-axis
(D) 16 along positive X-axis

Open solution

Q50. AIPMT 2011 Prelims

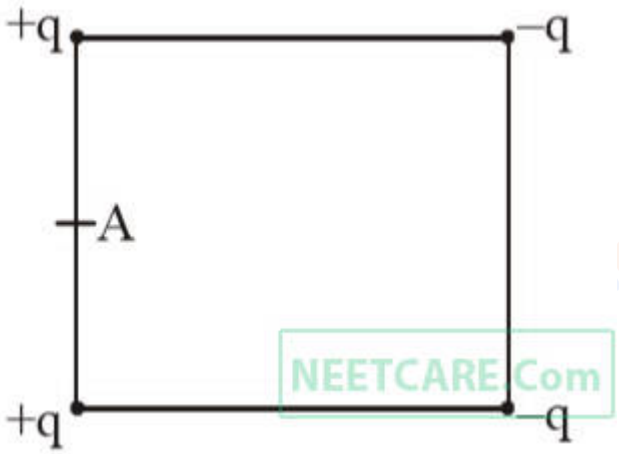
A charge Q is enclosed by a Gaussian spherical surface of radius R. If the radius is doubled, then the outward electric flux will

(A) increase four times
(B) be reduced to half
(C) remain the same
(D) be doubled

Open solution

Q51. AIPMT 2011 Prelims

Four electric charges +q, +q, − q and − q are placed at the corners of a square of side 2L (see figure). The electric potential at point A, midway between the two charges + q and +q, is



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(A) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{L}\left(1+\sqrt{5}\right)$
(B) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{L}\left(1+\frac{1}{\sqrt{5}}\right)$
(C) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{L}\left(1-\frac{1}{\sqrt{5}}\right)$
(D) zero

Open solution

Q52. AIPMT 2010 Mains

The electric field at a distance $\frac{3R}{2}$ from the centre of a charged conducting spherical shell of radius R is E. The electric field at a distance $\frac{R}{2}$ from the centre of the sphere is

(A) zero (B) E (C) $\frac{E}{2}$ (D) $\frac{E}{3}$

Open solution

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Q53. AIPMT 2010 Prelims

Two positive ions, each carrying a charge q, are separated by a distance d. If F is the force of repulsion between the ions, the number of electrons missing from each ion will be (e being the charge on an electron)

Open solution

(A) $\frac{4\pi\varepsilon_0 Fd^2}{e^2}$

(B) $\sqrt{\frac{4\pi\varepsilon_0 Fe^2}{d^2}}$

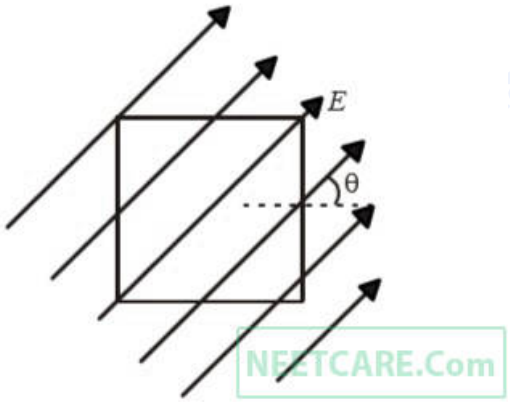
(C) $\sqrt{\frac{4\pi\varepsilon_0 Fd^2}{e^2}}$

(D) $\frac{4\pi\varepsilon_0 Fd^2}{q^2}$

Q54. AIPMT 2010 Prelims

A square surface of side L meter in the plane of the paper is placed in a uniform electric field E(volt/m) acting along the same plane at an angle θ with the horizontal side of the square as shown in figurre.

Open solution



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The electric flux linked to the surface, in units of volt m is

(A) EL^2

(B) $EL^2\cos\theta$

(C) $EL^2\sin\theta$

(D) zero

Q55. AIPMT 2009

Three concentric spherical shells have radii a, b and c (a < b < c) and have surface charge densities σ, $-\sigma$ and σ respectively. If V_A, V_B and V_C denote the potentials of the three shells, then, for c = a + b, we have

Open solution

(A) $V_C = V_B \neq V_A$

(B) $V_C \neq V_B \neq V_A$

(C) $V_C = V_B = V_A$

(D) $V_C = V_A \neq V_B$

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Q56. AIPMT 2009

The electric potential at a point (x, y, z) is given by V = $-x^2y - xz^3 + 4$

The electric field at that point is

(A) $\vec{E} = \hat{i}2xy + \hat{j}\left(x^2 + y^2\right) + \hat{k}\left(3xz - y^2\right)$

(B) $\vec{E} = \hat{i}z^3 + \hat{j}xyz + \hat{k}z^2$

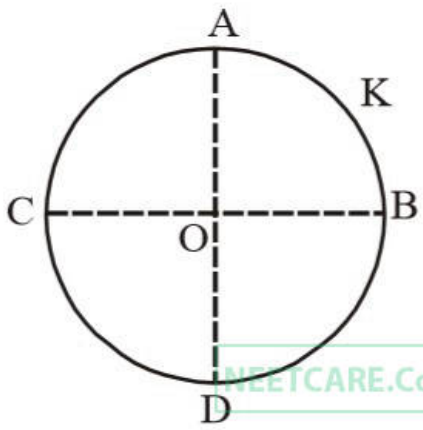
(C) $\vec{E} = \hat{i}\left(2xy - z^3\right) + \hat{j}xy^2 + \hat{k}3z^2x$

(D) $\vec{E} = \hat{i}\left(2xy + z^3\right) + \hat{j}x^2 + \hat{k}3xz^2$

Open solution

Q57. AIPMT 2008

A thin conducting ring of radius R is given a charge +Q. The electric field at the centre O of the ring due to the charge on the part AKB of the ring is E. The electric field at the centre due to the charge on the part ACDB of the ring is



(A) E along KO (B) 3E along OK (C) 3E along KO (D) E along OK

Open solution

Q58. AIPMT 2008

The electric potential at a point in free space due to charge Q coulomb is Q $\times 10^{11}$ volts. The electric field at that point is

(A) $4\pi\varepsilon_0 Q \times 10^{20}$ volt/m (B) $12\pi_0$Q $\times 10^{22}$ volt/m

(C) $4\pi\varepsilon_0 Q \times 10^{22}$ volt/m (D) $12\pi\varepsilon_0 Q \times 10^{20}$ volt/m

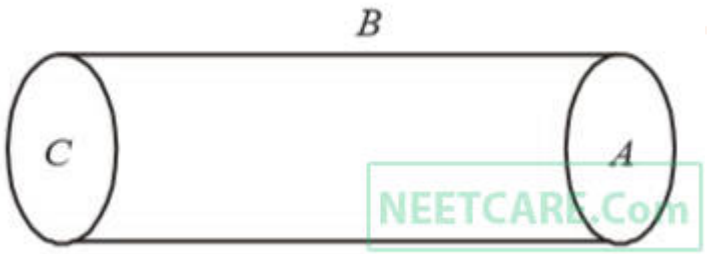
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Q59. AIPMT 2007

A hollow cylinder has a charge q coulomb within it. If ϕ is the electric flux in units of volt-metre associated with the curved surface B, the flux linked with the plane surface A in units of V-m will be

Open solution



(A) $\frac{q}{2\varepsilon_0}$

(B) $\frac{\phi}{3}$

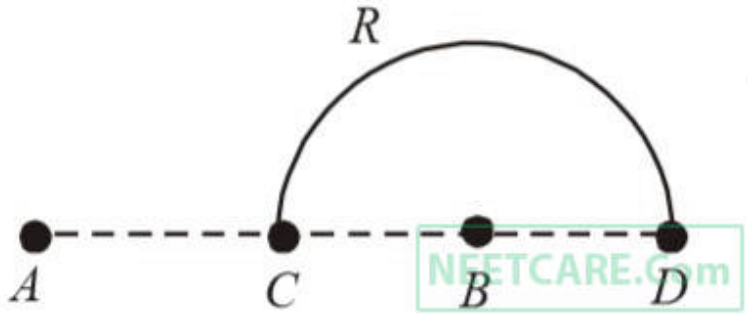
(C) $\frac{q}{\varepsilon_0} - \phi$

(D) $\frac{1}{2}\left(\frac{q}{\varepsilon_0} - \phi\right)$

Q60. AIPMT 2007

Charges +q and $-$q are placed at points A and B respectively which are a distance 2L apart, C is the midpoint between A and B. The work done in moving a charge + Q along the semicircle CRD is

Open solution



(A) $\frac{qQ}{2\pi\varepsilon_0 L}$

(B) $\frac{qQ}{6\pi\varepsilon_0 L}$

(C) $-\frac{qQ}{6\pi\varepsilon_0 L}$

(D) $\frac{qQ}{4\pi\varepsilon_0 L}$

Q61. AIPMT 2007

Three point charges +q, $-$ 2q and + q are placed at points (x = 0, y = a, z = 0), (x = 0, y = 0, z = 0) and (x = a, y = 0, z = 0) respectively. The magnitude and direction of the electric dipole moment vector of this charge assembly are

Open solution

(A) $\sqrt{2}qa$ along the line joining points (x = 0, y = 0, z = 0) and (x = a, y = a, z = 0)

(B) qa along the line joining points (x = 0, y = 0, z = 0) and (x = a, y = a, z = 0)

(C) $\sqrt{2}qa$ along +x direction

(D) $\sqrt{2}qa$ along +y direction.

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Q62. AIPMT 2006

An electric dipole of moment $\vec{p}$ is lying along a uniform electric field $\vec{E}$. The work done in rotating the dipole by 90° is

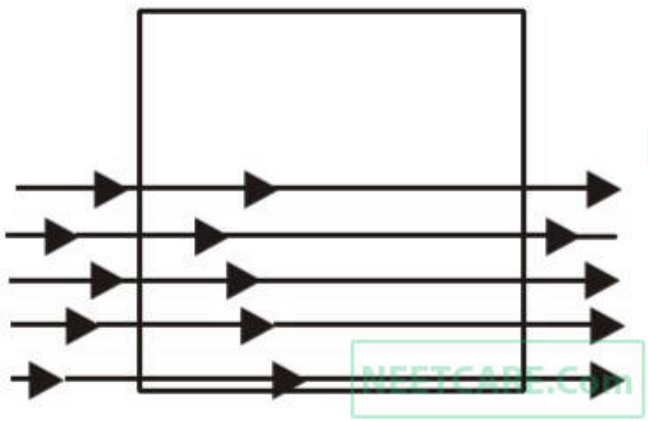
Open solution

(A) pE (B) $\sqrt{2}pE$ (C) pE/2 (D) 2pE

Q63. AIPMT 2006

A square surface of side L metres is in the plane of the paper. A uniform electric field $\vec{E}$ (volt/m), also in the plane of the paper is limited only to the lower half of the square surface (see figure). The electric flux in SI inits associated with the surface is

Open solution



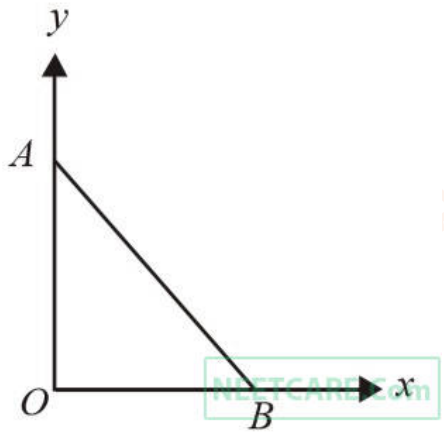
(A) EL^2 (B) $EL^2/2\varepsilon_0$ (C) $EL^2/2$ (D) zero

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Q64. AIPMT 2005

As per the diagram a point charge +q is placed at the origin O. Work done in taking another point charge $-Q$ from the point A [coordinates $(0, a)$] to another point B

Open solution



(A) zero

(B) $\left(\frac{qQ}{4\pi\varepsilon_0}\frac{1}{a^2}\right).\sqrt{2}a$

(C) $\left(\frac{-qQ}{4\pi\varepsilon_0}\frac{1}{a^2}\right).\sqrt{2}a$

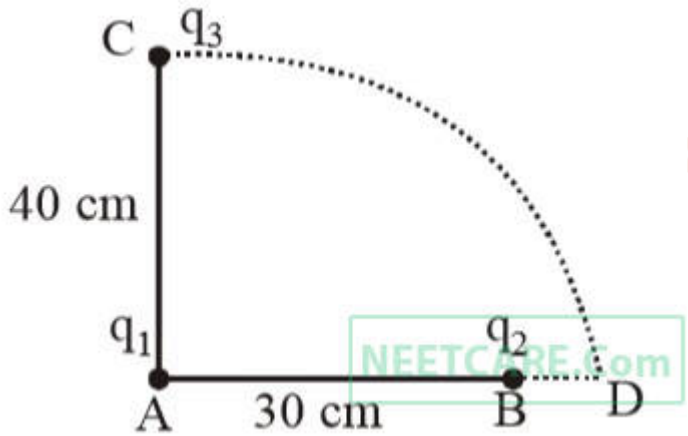
(D) $\left(\frac{qQ}{4\pi\varepsilon_0}\frac{1}{a^2}\right).\frac{a}{\sqrt{2}}$

Q65. AIPMT 2005

Two charges q_1 and q_2 are placed 30 cm apart, as shown in the figure. A third charge q_3 is moved along the arc of a circle of radius 40 cm from C to D.

Open solution

The change in the potential energy of the system is $\frac{q_3}{4\pi\varepsilon_0}k$ where k is



(A) $8q_1$ (B) $6q_1$ (C) $8q_2$ (D) $6q_2$

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Q66. AIPMT 2004

A bullet of mass 2 g is having a charge of 2 μC. Through what potential difference must it be accelerated, starting from rest, to acquire a speed of 10 m/s ?

(A) 5 kV (B) 50 kV (C) 5 V (D) 50 V

Open solution

Q67. AIPMT 2004

An electric dipole has the magnitude of its charge as q and its dipole moment is p. It is placed in a uniform electric field E. If its dipole moment is along the direction of the field, the force on it and its potential energy are respectively

(A) 2q.E and minimum
(B) q.E and p.E
(C) zero and minimum
(D) q.E and maximum

Open solution

Q68. AIPMT 2003

A charge q is located at the centre of a cube. The electric flux through any face is

(A) $\frac{2\pi q}{6(4\pi\varepsilon_0)}$
(B) $\frac{4\pi q}{6(4\pi\varepsilon_0)}$
(C) $\frac{\pi q}{6(4\pi\varepsilon_0)}$
(D) $\frac{q}{6(4\pi\varepsilon_0)}$

Open solution

Q69. AIPMT 2002

Identical charges (−q) are placed at each corners of cube of side b then electrostatic potential energy of charge (+q) which is placed at centre of cube will be

(A) $\frac{-4\sqrt{2}q^2}{\pi\varepsilon_0 b}$
(B) $\frac{-8\sqrt{2}q^2}{\pi\varepsilon_0 b}$
(C) $\frac{-4\,q^2}{\sqrt{3}\,\pi\varepsilon_0 b}$
(D) $\frac{8\sqrt{2}\,q^2}{4\,\pi\varepsilon_0 b}$

Open solution

Q70. AIPMT 2002

Some charge is being given to a conductor. Then its potential is

(A) maximum at surface
(B) maximum at centre
(C) remain same throughout the conductor
(D) maximum somewhere between surface and centre.

Open solution

Q71. AIPMT 2001

A dipole of dipole moment $\vec{p}$ is placed in uniform electric field $\vec{E}$ then torque acting on it is given by

(A) $\vec{\tau}=\vec{p}.\vec{E}$
(B) $\vec{\tau}=\vec{p}\times\vec{E}$
(C) $\vec{\tau}=\vec{p}+\vec{E}$
(D) $\vec{\tau}=\vec{p}-\vec{E}$

Open solution

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Q72. AIPMT 2001

A charge QμC is placed at the centre of a cube, the flux coming out from each face will be

(A) $\frac{Q}{6\varepsilon_0} \times 10^{-6}$ (B) $\frac{Q}{6\varepsilon_0} \times 10^{-3}$

(C) $\frac{Q}{24\varepsilon_0}$ (D) $\frac{Q}{8\varepsilon_0}$

Open solution

Q73. AIPMT 2000

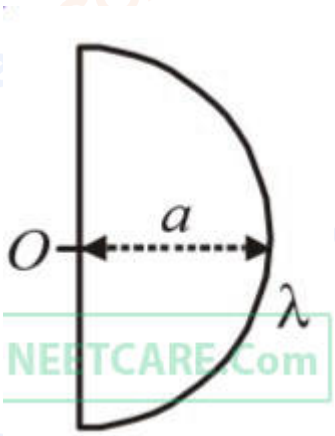
A charge Q is situated at the corner of a cube, the electric flux passed through all the six faces of the cube is

(A) $\frac{Q}{6\varepsilon_0}$ (B) $\frac{Q}{8\varepsilon_0}$ (C) $\frac{Q}{\varepsilon_0}$ (D) $\frac{Q}{2\varepsilon_0}$

Open solution

Q74. AIPMT 2000

Electric field at centre O of semicircle of radius a having linear charge density λ given as



(A) $\frac{2\lambda}{\varepsilon_0 a}$ (B) $\frac{\lambda\pi}{\varepsilon_0 a}$

(C) $\frac{\lambda}{2\pi\varepsilon_0 a}$ (D) $\frac{\lambda}{\pi\varepsilon_0 a}$

Open solution

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Answer Key

1. **A**	2. **D**	3. **C**	4. **C**	5. **D**
6. **B**	7. **D**	8. **D**	9. **D**	10. **C**
11. **B**	12. **D**	13. **C**	14. **C**	15. **D**
16. **D**	17. **B**	18. **C**	19. **B**	20. **B**
21. **A**	22. **C**	23. **C**	24. **A**	25. **A**
26. **D**	27. **A**	28. **A**	29. **C**	30. **C**
31. **A**	32. **B**	33. **A**	34. **B**	35. **A**
36. **D**	37. **A**	38. **B**	39. **D**	40. **B**
41. **A**	42. **D**	43. **D**	44. **B**	45. **A**
46. **A**	47. **B**	48. **D**	49. **A**	50. **C**
51. **C**	52. **A**	53. **C**	54. **D**	55. **D**
56. **D**	57. **D**	58. **C**	59. **D**	60. **C**
61. **A**	62. **A**	63. **D**	64. **A**	65. **C**
66. **B**	67. **C**	68. **B**	69. **C**	70. **C**
71. **B**	72. **A**	73. **B**	74. **C**	

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